



Advancing agriculture through IoT, Big Data, and AI: A review of smart technologies enabling sustainability

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ABSTRACT

This review addresses a critical aspect of modern agriculture: integrating the Internet of Things (IoT), Big Data, and Artificial Intelligence (AI) technologies to monitor and mitigate agricultural carbon emissions. We focus on the role of these advanced technologies in enhancing Climate-Smart Agriculture (CSA) and promoting sustainable farming practices. The paper provides a comprehensive review of how IoT, Big Data, and AI can be combined to monitor carbon footprints and support broader sustainability objectives in agriculture. As a key contribution, we propose a feasible, end-to-end system architecture tailored to the assessment of carbon footprint, combining IoT-enabled sensing, real-time data analytics, and predictive modeling. This study highlights the tangible benefits of these technologies through real-world case studies and evaluates their effectiveness in improving emission monitoring, operational efficiency, and environmental compliance. Furthermore, challenges such as data interoperability, device energy efficiency, and implementation costs are critically analyzed, providing insights into existing research gaps. The paper also identifies future directions, including scalable IoT-based carbon markets, Machine Learning (ML) algorithms for precision agriculture, and blockchain solutions for transparent carbon credit trading. The goal is to offer actionable insights into the adoption of cutting-edge technologies to achieve carbon neutrality and environmental sustainability in the agriculture sector.

1. Introduction

Global warming, an indisputable scientific fact indicated by decades of meteorological observations [148], is primarily driven by greenhouse gas (GHG) emissions resulting from human activities. These emissions trap heat in the atmosphere, leading to a rise in global average temperature and, consequently, global climate change [230,260]. Global temperatures have increased by approximately 1.1 °C over the past 150 years, with a significant portion of this warming occurring in the last 30 years [174]. This trend has contributed to more extreme weather events, deadly heatwaves, and severe drought, posing serious threats to agricultural production [198]. Carbon footprint monitoring has emerged as a vital approach to quantify and mitigate emissions, particularly in agriculture, which is a significant contributor to global GHG emissions [241]. Agriculture directly contributes approximately 8.5% of global GHG emissions through activities like enteric fermentation, rice production, and soil management [55]. An additional 14.5% of emissions result from land-use changes, primarily deforestation for agricultural expansion in developing regions [55]. These emissions are categorized as direct under Land-Use, Land-Use Change, and Forestry (LULUCF), while

indirect land-use change (iLUC) refers to emissions from diverting crops to non-food uses, such as biofuel production [86]. As cited by the United States Department of Agriculture, the agricultural sector alone already contributes over 10% of U.S. GHG emissions, a figure predicted to reach to 32% by 2050 if current trends persist [82,83].

Agriculture significantly contributes to global GHG emissions through various mechanisms. In the United States, as of 2022, carbon dioxide (CO₂) accounted for the largest share of greenhouse gas emissions at 79.7%, followed by methane (CH₄) at 11.1%, nitrous oxide (N₂O) at 6.1%, and other gases comprising 3.1% [246]. CO₂ emissions are primarily driven by the decomposition of soil organic matter and the open burning of crop residues, along with the use of fossil fuels in agricultural machinery, such as tractors and harvesters, and energy-intensive processes involved in processing and transporting agricultural products [147,16]. Methane (CH₄), another potent GHG with a global warming potential 28 times that of CO₂, is released in significant quantities from anaerobic conditions in flooded rice paddies, which are responsible for 10–12% of global CH₄ emissions [243]. Enteric fermentation in livestock and the decomposition of manure in moist environments also contribute to methane emissions [88,53]. Nitrous oxide (N₂O), with a

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global warming potential 298 times that of CO₂, primarily emanates from nitrogen-enriched soils due to synthetic fertilizers and animal manure applications [23]. Additionally, land use changes like deforestation for agricultural expansion exacerbate emissions by reducing carbon sequestration capacity and releasing stored carbon from forest ecosystems [93]. Given that soils are major carbon sinks, holding nearly three times more carbon than aboveground biomass and twice the amount found in the atmosphere, preserving soil health is crucial for balancing agricultural productivity and environmental sustainability [14]. This underscores the urgent need for sustainable agricultural practices to mitigate emissions and maintain the global carbon cycle [1]. Monitoring agricultural carbon footprints using advanced technologies such as IoT, Big Data, and AI is essential for achieving more sustainable agricultural practices and reducing overall emissions.

Addressing the multifaceted challenges posed by agriculture in the context of climate change, the concept of Climate-Smart Agriculture (CSA) has gained prominence [234]. CSA aims to transform agricultural development, fostering strong linkages among stakeholders at global, national, and local levels [158]. Its objective is a quadruple win: enhancing agricultural production and income, improving adaptability to climate change, minimizing GHG emissions, and bolstering national food security along with sustainable development goals [158,2]. Parallel to the evolution of CSA, Agriculture 4.0 represents the latest phase in agricultural advancement, leveraging technologies such as the IoAT, AI, Big Data analytics, robotics, and precision agriculture [227,122]. Technological integration is expected to be pivotal in addressing global food production challenges, ensuring resource sustainability, and adapting to the pressures of climate change. Agriculture 4.0, aligning with the principles of CSA, claims to be a promising route towards sustainable farming practices and carbon neutrality in agriculture, thereby contributing significantly to environmental governance and the mitigation of climate change impacts.

While existing reviews, such as those by [244], [28], and [217], have explored various methods of CSA and identified key challenges, they often overlook the evolving landscape of Agriculture 4.0 and beyond. Specifically, there needs to be more discussion regarding developing and applying a dedicated and end-to-end system architecture for precise carbon footprint monitoring. This review addresses this gap by focusing on the comprehensive integration of CSA, IoT, and Big Data into agricultural practices. Despite significant advancements in these fields over the past decade, which marked a transformative phase in agriculture, more holistic studies that effectively combine these technologies for efficient environmental monitoring and governance still need to be evaluated. This review will discuss this gap by:

- **Analyzing the Integration of Technologies in CSA:** Exploring how IoT, Big Data, and AI can be effectively integrated into CSA practices to enhance their efficiency and sustainability.
- **Evaluating Environmental Monitoring Based on Big Data:** Assessing the role of Big Data in environmental monitoring, particularly in the context of carbon footprint monitoring in agriculture. This includes examining the capacity of Big Data to provide actionable insights for policy-making and sustainable practice implementation.
- **Developing End-to-End Dedicated System Architectures:** Proposing or evaluating comprehensive system architectures that leverage IoT, Big Data, and ML for end-to-end carbon footprint monitoring in agriculture. This involves not just the collection and analysis of data but also the translation of these insights into practical, on-ground actions and policy recommendations.
- **Identifying Challenges and Opportunities:** Highlighting the challenges in deploying these technologies in a real-world agricultural context and suggesting potential solutions or areas for further research.

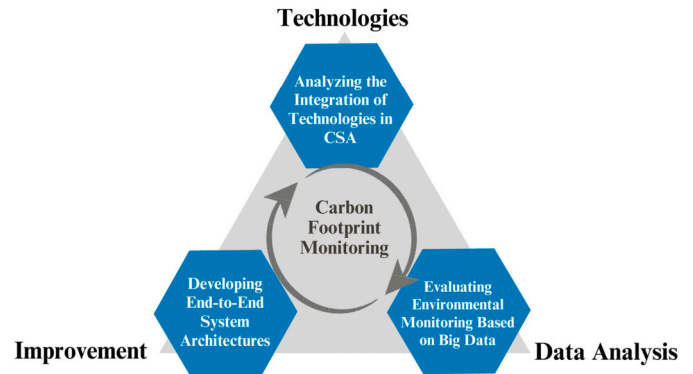


Fig. 1. Carbon Footprint Monitoring and the Key Objectives of this Study: Improvements in Carbon Footprint Auditing Techniques for Climate-Smart Agriculture (CSA).

Fig. 1 shows the overview process of carbon footprint monitoring and the key objectives of this review. By addressing these aspects, the aim of the paper is to provide a holistic understanding of the current state and potential future developments at the intersection of CSA, IoT, Big Data, and AI for environmental governance and sustainable agricultural practices.

This paper is organized into seven main sections following this introduction. Section 2 highlights various sources of carbon emission in agriculture. Section 3 examines the integration of IoT and Big Data in enhancing CSA. Section 4 evaluates the role of Big Data in environmental monitoring, with a focus on agricultural carbon footprint monitoring. Section 5 explores the development of end-to-end system architectures that leverage IoT, Big Data, and AI for effective carbon footprint monitoring. Section 6 discusses the challenges and opportunities in applying these technologies in agriculture. The paper concludes in Section 7.

2. Sources of greenhouse gases in agriculture

Agriculture is a significant contributor to greenhouse gas (GHG) emissions through various sources and processes, which vary across direct and indirect pathways. This section outlines these sources in accordance with the framework established by the GHG Protocol, categorizing emissions into three scopes: Scope 1 (direct emissions from sources owned or controlled by the entity), Scope 2 (indirect emissions from purchased energy), and Scope 3 (all other indirect emissions occurring across the value chain). This standardized classification aids in systematically analyzing and managing the carbon footprint of agricultural activities [211].

2.1. Calculation of carbon footprints

Accurately calculating carbon footprints in agriculture is essential for understanding and mitigating GHG emissions. This calculation follows a life cycle approach that integrates GHG emissions from the entire agricultural value chain, in line with the GHG Protocol guidelines [211,113].

Scope 1 includes direct emissions originating from on-site agricultural activities, such as fuel combustion in machinery, enteric fermentation from livestock, and methane (CH₄) emissions from rice paddies. Additionally, nitrous oxide (N₂O) emissions arise from the application of synthetic and organic nitrogen-based fertilizers. This scope includes direct N₂O emissions from nitrification and denitrification in soils, as well as indirect emissions from nitrogen volatilization and leaching [115]. These emissions are typically measured through emission factors or direct monitoring. Scope 2 accounts for emissions from purchased electricity, steam, and heating. Within agriculture, this includes electricity used for irrigation, grain drying, and greenhouse operations. Emission estimates are derived by calculating energy consumption and applying

Table 1
Consolidated Sources and Impacts of Greenhouse Gas Emissions in Agriculture.

Emission Source	Scope 1 (Direct Emissions)	Scope 2 (Indirect Emissions)	Scope 3 (Other Indirect Emissions)
CO₂ Emissions			
Soil Organic Matter Decomposition [203,87,64]	Natural decomposition of organic matter	Changes in land use and management practices	Ecosystem services, social, and economic impacts
Crop Residue Burning [201,235,137]	Direct emissions from burning residues	Transport and management of residues	Air quality, health impacts, and sustainability
Fossil Fuel Usage in Machinery [153,157,133]	Emissions from fuel combustion in machinery	Lifecycle emissions from fuel extraction and production	Socio-economic and ecological impacts of fossil fuel reliance
Energy-Intensive Processes [78]	Direct emissions from energy consumption	Indirect emissions from energy production and transport	Carbon footprint and sustainability impacts
CH₄ Emissions			
Anaerobic Conditions in Soils [178,84,73]	Direct CH ₄ release under flooded conditions	Land-use changes affecting emissions	Long-term ecological and economic impacts
Enteric Fermentation in Livestock [46,231]	Methane emissions during animal digestion	Emissions related to feed production and transport	Ecosystem and health impacts linked to livestock emissions
Manure and Crop Residue Decay [160,54]	Methane release from decomposition in moist conditions	Emissions from manure management practices	Impacts on air quality and soil health
N₂O Emissions			
Synthetic Fertilizer Usage [229,162,232]	Direct N ₂ O emissions from fertilizer application	Indirect emissions from fertilizer production and application methods	Broader ecological impacts, such as soil degradation and pollution
Animal Manure and Compost [135,208]	N ₂ O release from manure and compost applications	Indirect effects of manure management practices	Impacts on soil health, biodiversity, and sustainability
Agricultural Soil Management [62,61]	Direct emissions from nitrification and denitrification	Changes in management affecting leaching and runoff	Land-use sustainability and ecosystem outcomes

emission factors specific to the energy mix [114]. Moreover, Scope 3 encompasses indirect emissions along the entire agricultural value chain, such as emissions from producing agricultural inputs like fertilizers and pesticides, transporting goods, and managing waste. This scope includes critical processes like fertilizer manufacturing and the decomposition of crop residues, emphasizing the importance of life-cycle assessments to capture emissions holistically [113]. Furthermore, indirect N₂O emissions due to volatilization and nitrogen leaching from soils, as well as emissions from manure storage and decomposition, are integrated into Scope 3 assessments [115]. The carbon footprint calculation integrates GHG emissions by converting them into CO₂ equivalents using their respective Global Warming Potential (GWP) factors, as recommended by the IPCC [115]. The general equation is expressed as:

$$\text{Carbon Footprint (kg CO}_2\text{-e ha}^{-1}\text{)} = \sum (\text{Emissions (kg GHG)} \times \text{GWP factor}) \quad (1)$$

GHG emissions, including CO₂, CH₄, and N₂O, are converted to CO₂ equivalents using GWP factors to ensure uniformity in assessment. This approach encompasses both energy-related processes, such as fuel combustion and purchased electricity, and non-energy-related sources, including CH₄ emissions from enteric fermentation and manure decomposition, as well as N₂O emissions from nitrogen applications and soil nitrogen cycles. By incorporating upstream activities (e.g., fertilizer production) and downstream processes (e.g., transport and waste management), the life cycle approach provides a holistic and comprehensive evaluation of emissions across the agricultural production chain. Table 1 provides a comprehensive summary of the key GHG sources, categorized under Scope 1, Scope 2, and Scope 3, detailing their respective direct and indirect contributions to the overall carbon footprint.

2.2. Scope 1: direct emissions

Scope 1 emissions in agriculture represent direct, on-site emissions resulting from biological and energy-related processes, which contribute significantly to the agricultural carbon footprint. These emissions include the release of CO₂, CH₄, and N₂O through activities such as decomposition of organic matter in the soil, burning of crop residues,

livestock activities, fuel combustion in machinery and changes in land use.

The decomposition of soil organic matter is a primary source of biogenic CO₂ emissions, driven by microbial activity that varies with soil temperature, moisture, and organic content [110,203]. Intensive agricultural practices, such as excessive tillage and organic matter depletion, can accelerate this natural process, releasing excess CO₂ and disrupting the soil carbon balance [115]. Similarly, land use changes, particularly deforestation and the conversion of forests to agricultural land, result in the immediate release of CO₂ and other greenhouse gases through biomass removal and soil disturbances [257,165].

Crop residue burning is another major source of direct emissions, releasing both biogenic CO₂ and non-CO₂ gases, including CH₄ and N₂O [201]. Although the biogenic CO₂ from crop residue burning is typically excluded from direct carbon accounting under the GHG Protocol, the associated CH₄ and N₂O emissions are categorized under Scope 1 due to their significant impact on global warming [113]. Common practices such as burning rice and wheat residues contribute to these emissions, causing local air pollution, biodiversity loss, and soil degradation. Methane (CH₄) emissions primarily arise from enteric fermentation in ruminant animals and anaerobic conditions in paddy fields and manure decomposition sites [46,178]. During digestion, livestock release methane through belching, while anaerobic environments in rice paddies promote CH₄ release from decomposing organic material. Similarly, manure stored under moist conditions emits substantial amounts of CH₄ and N₂O as it decomposes [160].

Direct emissions of nitrous oxide (N₂O) are predominantly associated with the application of nitrogen-based inputs, such as synthetic fertilizers, animal manure, and compost [229]. Nitrification and denitrification processes in nitrogen-rich soils, especially under waterlogged conditions such as rice paddies, further exacerbate N₂O emissions [69]. Over-application of nitrogenous fertilizers and poorly managed irrigation systems can lead to increased N₂O emissions, making effective management practices crucial for mitigation.

Fossil fuel consumption in agricultural machinery also contributes to direct CO₂ emissions, with equipment such as tractors, harvesters, and irrigation systems consuming diesel and gasoline during operations [233,153]. Similarly, energy-intensive processes, including refrigeration

and the transportation of perishable goods, emit CO₂ directly from fuel combustion, especially in refrigerated transport units [78,92]. Effective on-site monitoring and emission control measures can help reduce the significant carbon footprint associated with these direct emissions.

2.3. Scope 2: indirect emissions

Scope 2 emissions in agriculture arise from energy and resource consumption that indirectly contribute to greenhouse gas (GHG) emissions through land use practices, energy-intensive operations, and associated supply chains. These emissions are linked to processes like soil carbon decomposition, crop residue management, fuel refinement, and electricity consumption, as detailed in Table 1. Indirect emissions from soil organic matter decomposition occur when agricultural activities, such as tillage and irrigation, alter the natural decomposition rate of organic matter [203]. Changes in land use and management practices affect carbon pools in soils, indirectly increasing CO₂ emissions. The incorporation of models that account for variables like crop types, soil carbon stocks, and environmental conditions is critical for estimating these emissions accurately.

Emissions associated with crop residue burning extend beyond the immediate combustion of residues, as pre- and post-burning activities—such as residue transport, irrigation, and collection—consume energy and indirectly emit CO₂ and non-CO₂ gases, including CH₄ and N₂O [201]. Scope 2 covers emissions from these energy-intensive activities, which contribute to the overall carbon footprint of residue management. The use of fossil fuels in agricultural machinery not only results in direct emissions but also indirect emissions during the lifecycle of the fuel, including extraction, refining, transportation, and distribution [153]. These processes collectively add to the indirect CO₂ emissions under Scope 2, particularly in mechanized and high-input farming systems.

Energy-intensive processes such as refrigeration and long-distance transportation of perishable goods further amplify Scope 2 emissions [78]. The electricity used in refrigeration systems, which is often derived from fossil fuels, contributes indirectly to CO₂ emissions. Additionally, the production and maintenance of refrigeration infrastructure and equipment also add to the carbon footprint. Indirect CH₄ emissions stem from land use and agricultural management practices that influence methane production. For example, land-use changes affecting soil properties can exacerbate methane release from anaerobic conditions in soils and manure management [178]. These emissions are closely linked to practices such as altered crop rotations, water management, and residue incorporation.

N₂O emissions under Scope 2 are influenced by the production and application methods of synthetic fertilizers and organic inputs. The production of fertilizers involves energy-intensive industrial processes, which generate indirect emissions before they are applied in the field [229]. Changes in land management techniques, such as improved nitrogen-use efficiency and optimized application timing, can help mitigate these indirect emissions by reducing the need for excess fertilizer inputs. Furthermore, emissions related to livestock feed production and transport contribute indirectly to methane and nitrous oxide emissions from livestock systems [46]. Feed cultivation, processing, and transportation all involve activities that indirectly generate CH₄ and N₂O emissions due to energy usage in production chains and transportation networks. Scope 2 assessments of these emissions provide a comprehensive understanding of how agricultural supply chains influence overall GHG emissions.

2.4. Scope 3: other indirect emissions

Scope 3 emissions encompass a wide range of indirect sources across the entire agricultural production chain, including upstream activities like fertilizer production and downstream processes such as transport and waste management. These emissions extend beyond the boundaries

of farm operations and consider the cumulative impacts on the environment, public health, and socio-economic conditions.

For CO₂ emissions, Scope 3 integrates the effects of residue transport, fertilizer production, and energy-intensive agricultural processes on biodiversity and ecosystem services [78]. Crop residue burning, though primarily contributing non-CO₂ gases under direct emissions, has cascading effects on air quality, biodiversity loss, and soil degradation under Scope 3. These emissions result in increased healthcare costs due to respiratory diseases, reduced agricultural productivity, and regional economic impacts like decreased tourism due to smog and pollution [201]. Sustainable alternatives, such as composting and no-burn practices, can play a crucial role in mitigating these impacts while preserving soil fertility and air quality.

Scope 3 also considers the broader environmental and socio-economic consequences of methane (CH₄) emissions. Methane emissions from livestock, particularly through enteric fermentation and manure decomposition, affect ecosystems and community well-being by degrading local air quality and contributing to global warming [46]. Mitigation strategies such as dietary changes in livestock and improved manure management are crucial in reducing CH₄ emissions, ensuring sustainable productivity without compromising environmental stability [20]. Additionally, anaerobic conditions in flooded fields, such as paddy rice cultivation, lead to long-term methane emissions, impacting not only agricultural sustainability but also community health.

For N₂O emissions, Scope 3 assesses the impacts on food security, land use, and ecosystem services. Excessive application of nitrogen-based fertilizers often leads to soil acidification and nutrient leaching, affecting both soil health and downstream water systems. Indirect emissions from fertilizer production and distribution contribute to significant ecological consequences, such as biodiversity loss and water pollution [229,62]. Regulatory pressures stemming from N₂O emissions can further affect smallholder farmers, requiring investments in environmentally friendly technologies that may strain financial stability [135]. Policies aimed at supporting farmers through financial incentives and knowledge-sharing programs could help ease this burden while promoting sustainable practices.

Land use changes, including deforestation and agricultural expansion, also contribute to Scope 3 emissions by altering carbon sinks and increasing greenhouse gas concentrations [257]. Changes in soil carbon stocks and reduced vegetation cover diminish the ability of ecosystems to sequester carbon, exacerbating atmospheric GHG levels. Addressing these challenges involves adopting sustainable land management practices that integrate reforestation efforts and conservation agriculture.

Summary on Challenges and Future Research Scopes. In addressing the complexities of GHG emissions in agriculture, researchers confront a series of challenges that impede our understanding and management of these emissions. One of the primary challenges is achieving a micro and nano-level understanding of emissions, particularly in soil systems. The intricate interplay between soil microbes, organic matter, and farming practices has a profound impact on emission dynamics, yet this remains inadequately understood. Advancements in nanosensors and micro-level analysis techniques are crucial for delving deeper into these interactions. Moreover, the remote deployment and data collection from diverse agricultural terrains poses significant logistical challenges, necessitating the development of robust, energy-efficient sensors and reliable data transmission methods.

3. Analyzing the integration and architecture of technologies in CSA

Integrating modern technologies, such as IoT and Big Data, into CSA practices is critical to enhancing agricultural efficiency and sustainability. This section delves into how these technologies are integrated into CSA, explores their potential benefits, and presents case studies and success stories that highlight their practical applications in agriculture.

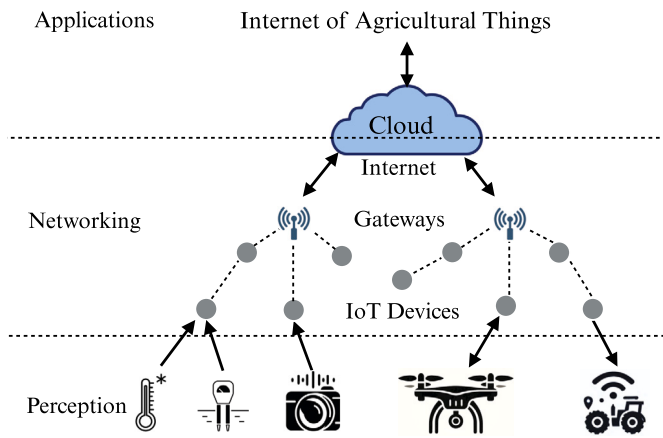


Fig. 2. Architecture of an Internet of Agricultural Things (IoAT).

3.1. Integration of IoT and Big Data in CSA

IoT and AI technologies have significantly advanced sustainable agriculture by optimizing resource utilization and improving environmental outcomes. IoT-enabled precision irrigation systems, as studied by Prabhat et al. [196], have demonstrated a 30% reduction in water use and improved crop yields by ensuring timely and precise irrigation based on real-time soil moisture and weather data. Similarly, AI-driven digital soil mapping algorithms, like those developed by Perennial, have reduced physical soil sampling requirements by 40%, improving the accuracy of carbon sequestration assessments and enabling targeted interventions for regenerative agricultural practices [161]. IoT-integrated energy-efficient irrigation systems have been shown to reduce energy consumption by up to 25%, as demonstrated in López-Morales et al.'s work on smart water wells [154]. Further, Big Data analytics in agriculture improved crop yield predictions by up to 25%, and optimized the use of nitrogen fertilizers, leading to cost reductions of approximately 15% [255].

3.1.1. Internet of things (IoT)

Architecture of IoT in Agriculture. Fig. 2 illustrates a comprehensive architecture of the Internet of Agricultural Things (IoAT), integrating modern technologies in agriculture. The IoAT builds upon the foundation of the IoT, enabling data-driven insights and automation in agricultural practices [181]. At the perception layer, sensors play a vital role, monitoring environmental parameters like soil moisture, temperature, humidity, and atmospheric conditions [222]. For example, soil moisture sensors provide real-time data, which helps in optimizing irrigation schedules. This layer also includes sensors for tracking greenhouse gas emissions, such as infrared gas analyzers for CO₂ and methane detection, Normalized Difference Vegetation Index (NDVI) sensors for crop health assessment, and weather stations for collecting microclimatic data. Together, these sensors create a detailed environmental profile necessary for precision agriculture. However, it's essential to critically assess the term 'smart' when referring to technologies like smart irrigation systems [259]. While they are designed to trigger precise irrigation based on sensor inputs, the true "smartness" of these systems depends on the accuracy of the underlying models and the thresholds, which must be thoughtfully defined through robust research and adaptation to local conditions. The integration of advanced sensing capabilities into the IoAT ensures a more robust and context-aware agricultural system capable of responding dynamically to real-time data.

Agricultural machinery, such as tractors, is increasingly equipped with sensors and actuators to enable automation. Global Positioning System (GPS)-guided tractors can now perform tasks like planting and fertilizing with a high degree of precision, utilizing GPS sensors and actuators. Drones further extend the reach of this technology, capturing aerial images and data across large fields, which aids in crop health

assessment [183,29]. At the networking layer, communication between these components is essential. However, referring to this communication as "seamless" can be misleading. Data from these sensors are transmitted to the cloud for further analysis, though this requires stable infrastructure, which might not always be available in all agricultural contexts. At the application layer, IoAT offers solutions such as precision agriculture, predictive maintenance, crop monitoring, and supply chain optimization. While this has the potential to revolutionize agriculture, particularly in well-resourced environments, the applicability of such systems varies based on local infrastructure and economic conditions. It's important to maintain a critical view of this revolution, which often assumes technology availability and ignores the challenges of implementation in under-resourced regions.

Integration of IoT in CSA. Integrating IoT into Climate-Smart Agriculture (CSA) represents a noteworthy advancement in agricultural practices. IoT encompasses a network of interconnected sensors and devices, which facilitate real-time data collection from various agricultural elements, such as fields, livestock, and equipment. However, it is critical to recognize that farmers have always collected data—albeit informally—and that IoT simply formalizes and digitizes this process. The value of IoT in CSA lies in its ability to provide more granular and accurate data, which can aid in optimizing resource use and improving decision-making processes [27]. Insights into soil health, moisture levels, and weather conditions allow for refined practices aimed at enhancing yield and sustainability. However, it's important to challenge the assumption that IoT "revolutionizes" farming practices in all contexts. The remote monitoring and control capabilities of IoT allow farmers to respond to challenges, such as climate variability, more effectively. However, IoT adoption remains limited in many regions due to infrastructural, financial, and educational barriers, and its benefits are not uniformly accessible. In this context, IoT's potential to drive climate-resilient and sustainable practices must be critically evaluated. The effectiveness of IoT depends on local adaptation, availability of robust models, and socio-economic conditions. Without addressing these fundamental challenges, the integration of IoT into agriculture risks exacerbating existing inequalities, where only the most well-resourced farmers can fully benefit from these technologies.

The integration of Artificial Intelligence of Things (AIoT) [236,263] in CSA enhances data processing and decision-making through advanced sensor technologies for real-time emission monitoring, particularly for gases like CO₂, CH₄, and N₂O. Technologies such as Non-Dispersive Infrared (NDIR) sensors [116], electrochemical and photoacoustic gas sensors [169], allow precise tracking of emissions from activities like fertilizer use and livestock management. This data, combined with AI-driven models, aids in emission reduction and contributes to sustainable farming practices. Moreover, while IoT systems monitor soil moisture, environmental, and micro-meteorological data, they do not directly measure plant parameters such as pH or temperature. Instead, AIoT leverages environmental sensors and predictive models to infer plant health by tracking factors like nutrient levels, temperature, and humidity, optimizing resource use for enhanced growth and sustainability.

Research and development in IoT-based CSA can be categorized into three primary approaches:

Data collection for informed farming practices, where IoT is utilized to gather crucial agricultural data through sensors measuring environmental and crop parameters such as temperature, humidity, pH value, leaf color, etc. [121,44,65,195].

Automation of farming practices, involving the use of wireless sensor and actuator networks for cloud-based decision-making and optimized irrigation systems [214,189]. This approach leverages various sensors and robots or actuators for automated farming operations, guided by principles of system visibility, safety, simplicity, feedback, extensibility, and cognitive load reduction [4,3].

In-network control for reducing power consumption and overhead while maintaining Quality of Service (QoS). This involves the optimization of resource utilization within the IoT network, in-

cluding communication channels and power sources. Innovations such as traffic pattern-aware resource allocation and AI-enabled energy-efficient networking for wireless sensor networks contribute to this effort [10,107,245]. However, the effectiveness of these solutions depends heavily on the accuracy of the models that underpin the data processing and integration. While they have the potential to significantly reduce power consumption and overhead, there is also the risk of errors or inefficiencies if the models are not robust or context-appropriate.

Electric and autonomous machinery. IoT sensors embedded in the machines offer real-time performance monitoring, including battery health and operational efficiency, which helps optimize their use and reduces reliance on fossil fuels [139,242]. However, ensuring that these sensors function effectively across diverse agricultural conditions presents significant hurdles, especially in regions with limited access to robust connectivity infrastructure. Additionally, the high cost of electric machinery and IoT sensor networks remains prohibitive for smallholder farmers, raising concerns about equitable access to these innovations. Big Data plays a role in analyzing machine performance, but processing and storing vast amounts of data collected from machinery in real time requires advanced computing infrastructure, which may not always be available in rural agricultural areas.

Solar-powered irrigation systems. IoT sensors monitor soil moisture and weather conditions to ensure irrigation only occurs when necessary, optimizing water and energy use [188]. While these systems hold significant potential for energy conservation, challenges arise in ensuring consistent performance in areas with variable sunlight or extreme weather. Furthermore, the initial costs of solar-powered systems and the data infrastructure required to monitor them may limit adoption. Big Data can provide insights by analyzing historical irrigation patterns and optimizing water use across different crop cycles. However, the complexity of analyzing such diverse datasets—especially when accounting for environmental variables—poses technical and resource-related challenges. Data security and privacy also become pressing issues, as the integration of IoT systems generates sensitive information related to agricultural productivity.

Scaling precision agriculture technologies. IoT-enabled equipment, such as GPS-guided tractors, can significantly reduce fuel consumption by optimizing routes and minimizing unnecessary field passes [224,249]. However, precision agriculture depends heavily on reliable data inputs and seamless communication between devices, which can be difficult to achieve in areas lacking sufficient network coverage. While Big Data analytics can refine these processes by offering actionable insights on fuel consumption patterns, the challenge lies in the continuous processing of high-volume data streams from multiple sources. This process requires extensive cloud-based infrastructure, and the costs associated with managing such data systems can be prohibitive for smaller farms. Additionally, the centralized control of such data by large tech companies raises ethical concerns regarding data ownership and farmers' autonomy in decision-making.

Each of these areas is critical to IoT applications in CSA, potentially leading to more effective, adaptable, and sustainable agricultural practices. However, future development and research must focus on improving the accuracy and reliability of these systems to minimize mistakes or misinterpretations. By refining these approaches, IoT can become a more reliable and impactful tool in agriculture.

3.1.2. Big Data analytics

Big Data represents a paradigm shift in data management, characterized by its three primary features: volume, velocity, and variety [209]. It refers to datasets that are too large, too fast, or too complex for traditional data processing techniques. In agriculture, Big Data encompasses vast amounts of data generated from diverse sources at an unprecedented speed, providing a wealth of information crucial for understanding and managing agricultural ecosystems.

Architecture of Big Data in Agriculture. The architecture of Big Data in agriculture is a comprehensive, multi-layered framework designed

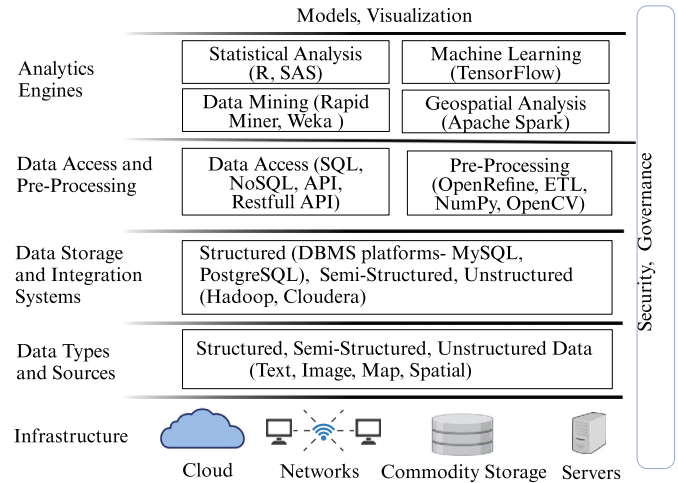


Fig. 3. Architecture of a Big Data Ecosystem for Smart Agriculture.

to efficiently manage and analyze vast and varied datasets, transforming them into actionable insights for improved agricultural practices [200,125,212]. Fig. 3 shows the overview architecture of a Big Data ecosystem for smart agriculture. At the base of this architecture lies the Infrastructure Layer, comprising Cloud platforms such as Amazon Web Services [56], Google Cloud Platform [142], and Microsoft Azure [57] that offer scalable storage and computing resources. This layer also includes essential network infrastructure for data transmission and servers that provide the physical backbone for data processing and storage. The Data Types and Sources Layer follows, encompassing Structured Data, typically stored in relational databases like MySQL [75] and PostgreSQL [74]; Semi-Structured Data, such as Extensible Markup Language (XML) [102] and JavaScript Object Notation (JSON) [193]; and Unstructured Data formats like text, images, maps, and spatial data [8]. This data is collected from various sources, including remote sensors, satellite imagery, weather stations, and GPS-enabled equipment, offering a comprehensive view of agricultural environments.

The subsequent layer, Data Storage and Integration Systems employs platforms like Hadoop [43] and Cloudera [163] to store and process both semi-structured and unstructured data. These systems are integral in consolidating disparate data into a unified format for advanced analysis. Following this is the Data Access and Pre-Processing Layer, where tools such as Structured Query Language (SQL) and NoSQL databases [52], along with Application Programming Interfaces (APIs), facilitate data access. Pre-processing tools like OpenRefine [248], Extract, Transform, Load (ETL) utilities [17], NumPy [182], and OpenCV [45] are utilized for data cleaning and transformation. The Analytics Engines Layer is where the heavy lifting of data analysis occurs, employing Statistical Analysis software (e.g., R [76], Statistical Analysis System (SAS) [206]), ML platforms (e.g., TensorFlow [186]), and Data Mining tools (e.g., RapidMiner [138], Weka [100]) to extract insights. Geospatial analysis, using tools like Apache Spark [215], plays a crucial role in this layer, particularly for spatial data relevant to agriculture. The final layer is Visualization and Model, where tools like Tableau [33] and Power BI [34] convert complex data analyses into understandable visual formats, and predictive models are developed to provide actionable recommendations. This top layer is key in decision-making and implementing optimized agricultural practices, culminating the data journey from collection to actionable insights.

Integration of Big Data in CSA. Big Data is increasingly recognized as a pivotal force in transforming agriculture into a more data-driven, efficient, and climate-resilient system. Its integration into Climate-Smart Agriculture (CSA) leverages various approaches to address different aspects of agricultural practices and management [176]. One such approach is adaptation to climate variability. For example, [37] utilized substantial volumes of climate data to aid in climate change adaptation

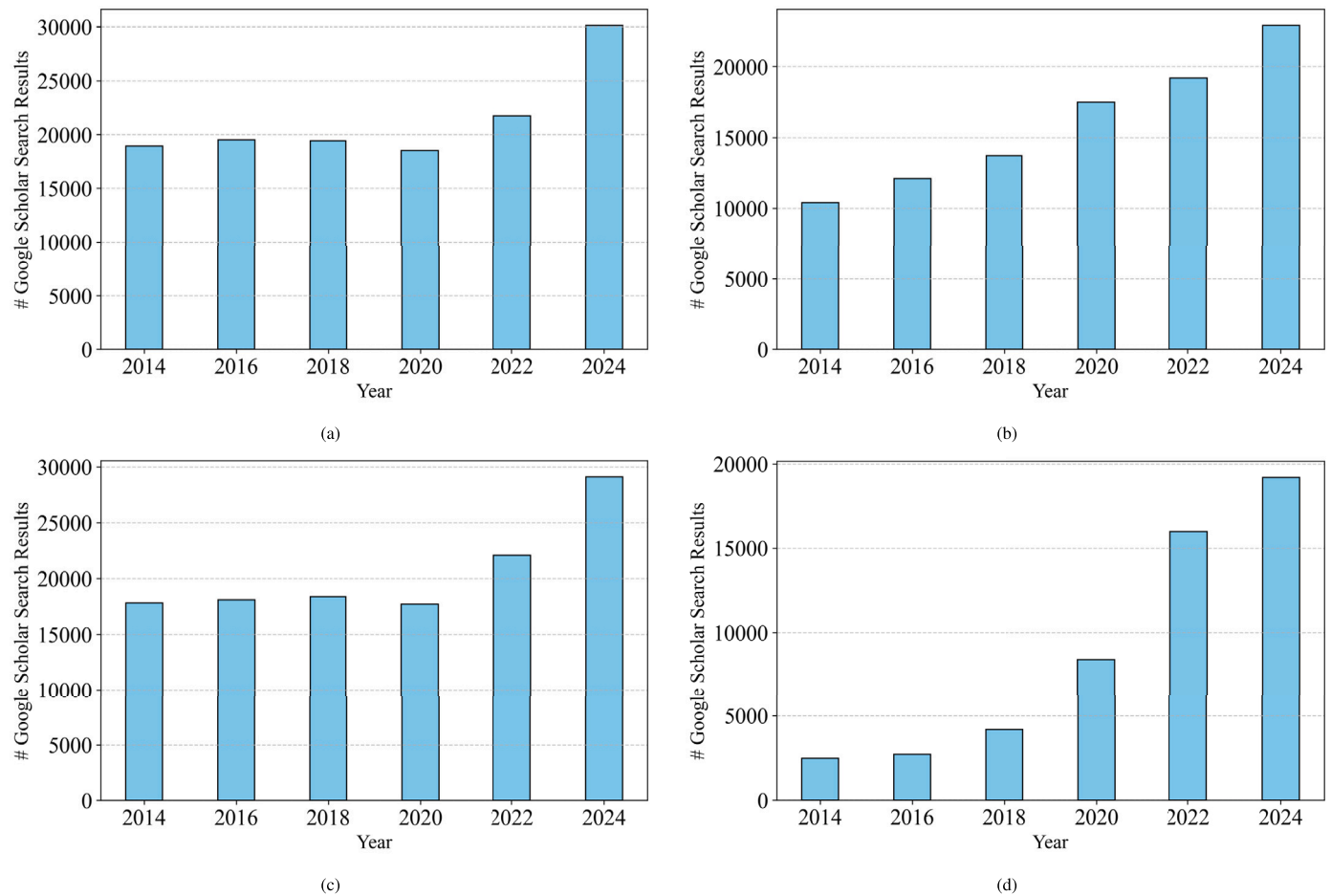


Fig. 4. Overview of the growing research focus on Big Data, IoT, Sensors, and AI/ML in Carbon Footprint Monitoring and Sustainable Agriculture based on Google Scholar search results: (a) Big Data, (b) sensor, (c) IoT, and (d) AI & ML.

by developing a web-based solution that offers maps depicting seasonal mean temperatures. Through real-time and historical data, Big Data analytics enable some farmers who use precision agriculture techniques to adapt their practices to changing climate conditions [125]. This capability is critical for reducing climate-related risks and supporting food security. However, it is essential to acknowledge that while Big Data holds significant promise, its benefits may not be equally accessible to all farmers. Smaller-scale farmers, particularly in developing regions, may face challenges in adopting these technologies due to barriers in infrastructure, cost, or technical expertise. Additionally, the centralization of data by technology companies and large corporations can shift decision-making power away from individual farmers, raising ethical concerns about data ownership, transparency, and the potential for exploitation by larger entities. Addressing these ethical dimensions is crucial to ensure that Big Data promotes equity and sustainability across the agricultural sector.

Beyond the realm of direct farming practices, Big Data also contributes significantly to supply chain optimization and policy development. It provides insights that extend beyond the confines of the farm, assisting in optimizing the entire agricultural supply chain, enhancing market access, and formulating effective agricultural policies. This broad application promotes a comprehensive and integrated approach to CSA [149]. Additionally, the field of Big Data in agriculture includes innovative methods for data integration, storage, processing, and visualization. These state-of-the-art solutions range from long-term analytics, which is essential for strategic planning, to real-time analytics that enables immediate decision-making [98]. An exemplary application of Big Data in commercial decision-making is seen with companies like Climate Corp [112] and John Deere [146]. They utilize localized

weather data to predict crop yields and accordingly adjust crop insurance premiums, analyzing an extensive dataset spanning over 60 years of weather patterns [24]. This diverse application of Big Data in CSA underscores its critical role in enhancing agricultural efficiency, resilience, and sustainability.

To illustrate the growing research trends in Big Data, IoT, Sensors, and AI/ML for Carbon Footprint Monitoring and Sustainable Agriculture, we conducted a Google Scholar search for each category using relevant keywords and filtered results based on publication year. The number of scholarly articles indexed over the years 2014–2024 was extracted to analyze research growth. Fig. 4 provides an overview of this increasing interest, reflecting how emerging technologies are shaping sustainability efforts in agriculture. Notably, while Big Data and IoT-based solutions have shown steady growth, research on AI and ML has experienced a significant surge in recent years. This rise can be attributed to advancements in deep learning models, federated learning, and edge computing, enabling more efficient carbon monitoring and predictive analytics. The application of AI/ML in agricultural sustainability and climate resilience is now a focal point, with increasing studies leveraging neural networks, reinforcement learning, and hybrid AI techniques to optimize resource use, emissions tracking, and precision agriculture. This trend highlights a broader shift toward data-driven, automated decision-making in mitigating agriculture's carbon footprint.

Summary on Challenges and Future Research Scopes. The integration of modern technologies such as IoT and Big Data into CSA has opened new opportunities for enhancing agricultural efficiency and sustainability. However, this integration presents several challenges and avenues for future research. One of the key challenges is the seamless integration of diverse technologies to ensure coherent and efficient func-

Table 2
Categorized Roles of Big Data in Agricultural Environmental Monitoring.

Aspect	Description	References
Real-Time Monitoring and Environmental Adaptation		
Real-Time Analysis	Immediate processing of environmental data for prompt decision-making, including dynamic responses to changing field conditions.	[11,109,99,264]
Climate Impact Studies	Analyzing and predicting the effects of climate variability to inform adaptive farming practices and enhance climate resilience.	[60,105,95]
Predictive Capabilities and Pest Control		
Pest and Disease Prediction	Utilizing historical and real-time data to predict pest infestations and outbreaks, allowing timely interventions to protect crop health.	[261,207,145,118]
Sustainable Resource Management and Optimization		
Resource Management	Optimizing the use of water, fertilizers, and other inputs to boost crop yields while minimizing environmental harm. This also includes enhancing operational efficiency.	[126,25,39,175]
Supply Chain Optimization	Enhancing logistics, storage, and distribution processes by analyzing production and market trends, ultimately reducing waste and aligning with market demands.	[155,6,171,26]

tioning. The vast and varied nature of data collected from IoT devices poses significant difficulties in terms of storage, processing, and analysis. Ensuring data accuracy, reliability, and timely processing despite such diversity and volume remains a critical area for continuous improvement. Additionally, developing robust and scalable IoT networks that can reliably operate in different agricultural environments, from high-tech farms to resource-limited rural areas, requires further exploration.

Beyond the technical challenges, there are substantial socio-economic considerations. Developing advanced Big Data analytics tools and predictive models to support decision-making is crucial, but so is ensuring that these innovations are accessible to all farmers, including those in remote and resource-constrained areas. To truly benefit all stakeholders, efforts must go beyond the creation of sophisticated technologies. Research must focus on developing low-cost, user-friendly solutions that are easy to implement even with limited infrastructure. This could involve creating adaptable IoT devices that can function with intermittent connectivity, or localized Big Data processing platforms that minimize reliance on expensive cloud services. Moreover, it is essential to invest in education and capacity-building programs to empower farmers, particularly in marginalized communities, to effectively use these technologies. The success of IoT and Big Data integration in agriculture depends not only on the availability of technology but also on ensuring that farmers have the knowledge and tools to harness its potential. This includes designing intuitive decision-support systems that can translate complex data insights into actionable guidance that farmers can easily understand and implement. Lastly, developing collaborative frameworks between governments, NGOs, and private-sector entities is critical to facilitate technology dissemination and provide financial support, training, and infrastructure development in underprivileged areas. Only by addressing these human-centered aspects will the technology truly be able to transform global agriculture and ensure that the benefits are distributed equitably across all regions, regardless of resources.

4. Evaluating environmental monitoring based on Big Data

Big Data has emerged as a pivotal tool in revolutionizing environmental monitoring within the agriculture sector, offering profound insights and capabilities in managing the ecological aspects of farming. This section evaluates the transformative impact of Big Data in agricultural practices, specifically focusing on its role in facilitating precise environmental assessments, predicting ecological changes, and enhancing resource management, all of which are crucial for reducing the carbon footprint of agricultural activities.

4.1. The role of Big Data in agricultural environmental monitoring

One significant facet of Big Data in this context is its role in enabling real-time analysis of crucial environmental conditions. By processing data immediately, it empowers farmers to make prompt decisions vital for crop health and yield. For example, studies conducted by Akanbi et al. [11] and Himeur et al. [109] have demonstrated how real-time analysis of soil and weather data can enhance irrigation strategies and pest control practices. Another pivotal application is in pest and disease prediction. Through the analysis of patterns and historical data, Big Data equips us to forecast pest infestations and disease outbreaks, enabling preemptive actions to protect crops and reduce potential losses. Notable research, such as [261] and [47], underscores how data analytics can foresee impending pest attacks, facilitating timely interventions. Further, Big Data plays a crucial role in optimizing resource management within agriculture. It assists in efficient utilization of resources like water, fertilizers, and pesticides, resulting in increased crop yields and minimized environmental impact [184]. Research by Kamyab et al. [126] illustrates how Big Data contributes to achieving efficient water usage through smart irrigation systems.

Moreover, Big Data aids in climate impact studies, enabling farmers to adapt their practices to changing environmental conditions by analyzing historical climate patterns and current trends. Insights from studies like those conducted by [60], [105], and [95] provide valuable guidance for climate resilience planning in agriculture. Big Data analytics extends beyond the farm, optimizing the agricultural supply chain. By analyzing market trends and logistics data, it ensures efficient distribution, minimizes waste, and aligns production with market demands. As demonstrated by Joash [155], Big Data has the potential to significantly enhance the sustainability of the agricultural supply chain.

In summary, the integration of Big Data in agricultural environmental monitoring is instrumental in transitioning farming practices toward a data-driven paradigm. This approach enhances efficiency, optimizes resource utilization, and supports adaptive strategies for climate resilience. By categorizing key applications such as real-time monitoring, predictive analytics, and sustainable resource management, we can systematically analyze how Big Data technologies drive agricultural sustainability and productivity (see Table 2).

4.2. Big Data for carbon footprint monitoring in agriculture

Monitoring and mitigating the carbon footprint of agricultural practices is vital in the quest for sustainable farming and environmental conservation. The integration of Big Data technologies into this realm has revolutionized the way we collect, analyze, and manage data related

Table 3
Categorized Roles of Big Data in Carbon Footprint Monitoring and Sustainable Agriculture.

Role	Description	References
Emission Monitoring and Analytics		
Emissions Tracking and Analysis	Continuous tracking of emissions from various farming activities using real-time data from sensors, satellite imagery, and IoT devices to identify emission hotspots.	[254,5,48,175]
Predictive Analytics for Emission Reduction	Employing predictive models to forecast emissions based on historical data, environmental conditions, and farming practices. This enables targeted interventions to reduce carbon emissions.	[80,226,210,252,104]
Resource and Practice Optimization		
Sustainable Farming Practices	Assessing environmental impacts of current farming methods and recommending eco-friendly, data-driven alternatives for long-term sustainability.	[124,36,120,111]
Policy and Climate Adaptation		
Policy Development and Compliance	Facilitating policy formulation by providing data-driven insights and ensuring farmers comply with emissions standards.	[254,177,42]
Enhanced Climate Change Adaptation	Analyzing long-term climate patterns to help farmers adapt their practices and improve resilience to changing weather conditions and environmental stressors.	[220,190,19,111]

to greenhouse gas (GHG) emissions, energy consumption, and resource efficiency in agriculture.

Big Data facilitates the detailed tracking and analysis of emissions arising from diverse farming activities [254]. It covers a wide range of agricultural processes, including tillage, fertilization, crop residue management, and livestock rearing [247]. By integrating data from sensors, satellite imagery, and IoT devices, Big Data systems provide critical insights into emission hotspots and patterns, allowing for targeted mitigation strategies [5]. For example, research by Zihan et al. [127] demonstrates the use of GPS-based Big Data, which is applicable for estimating fuel consumption and emissions from agricultural tractors and other vehicles as well. Similarly, Tullo et al. [80] highlights how Big Data is applied to monitor livestock emissions and develop effective mitigation measures. Furthermore, Big Data empowers predictive analytics for emission reduction by leveraging historical data, current farming practices, and environmental variables to forecast areas where carbon footprints can be minimized [226].

Further, Big Data enables resource use optimization in agriculture. Farmers can harness the power of data analytics to make informed decisions about the efficient utilization of resources like water, fertilizers, and energy, resulting in a reduced overall carbon footprint. For example, Khadijeh et al. [18] demonstrates how Big Data can optimize irrigation schedules, reducing water consumption while maintaining crop yield. Big Data also plays a pivotal role in advocating sustainable farming practices. It facilitates the assessment of the environmental impacts of existing farming methods and recommends eco-friendly alternatives. This includes practices like precision agriculture, reduced tillage, and integrated pest management. Research by [124] and [36] highlights the use of Big Data analytics to guide farmers in adopting sustainable practices that lower their carbon footprint. Lastly, Big Data contributes to enhanced climate change adaptation in agriculture [220]. By analyzing climate data and its implications for farming, Big Data enables the formulation of strategies to cope with shifting weather patterns, ensuring food security and sustainability.

Table 3 categorizes the various roles of Big Data in carbon footprint monitoring, highlighting its crucial contributions to emission tracking, predictive analytics, resource optimization, and policy support. The integration of Big Data in carbon footprint monitoring in agriculture represents a significant technological advancement, yet it should be viewed as only one part of a broader paradigm shift toward truly sustainable and responsible farming. While harnessing data analytics can contribute to eco-friendly and resource-efficient practices, the focus on carbon alone is limited. Carbon footprinting, while valuable, addresses only one dimension of sustainability, often neglecting crucial considerations such as ecological footprints, ethical impacts, and inter-generational consequences of farming practices. A more holistic approach that includes these dimensions would better align agricultural practices with long-

term environmental and social goals. Moreover, this reliance on Big Data could risk pushing agriculture further toward industrialization, which some argue may not be the optimal way forward for achieving sustainability. Critics contend that industrial agriculture often leads to increased environmental degradation, loss of biodiversity, and socio-economic disparities. Thus, while Big Data offers tools for improving efficiency and reducing emissions, it must be employed with caution, ensuring that technological solutions don't overshadow or undermine the diverse ethical, ecological, and cultural needs of global farming communities. In this light, the future of agriculture will depend on a balanced, critical approach to data-driven innovations, ensuring that technologies not only support carbon reduction but also foster broader ecological, ethical, and social sustainability. Moving forward, it is essential to cultivate a nuanced, inclusive vision for agriculture that questions the long-term impacts of industrialization and promotes practices that truly benefit both people and the planet.

4.3. Case studies on IoT and Big Data integration in carbon footprint monitoring and sustainability

The integration of IoT and Big Data technologies has led to successful implementations across various sectors, showcasing their transformative potential in reducing emissions and improving operational efficiency. For instance, the Pacific Gas and Electric Company (PG&E) in California utilized IoT sensors and Big Data analytics in a smart grid optimization project to enhance energy distribution and renewable energy integration. By forecasting energy demand and minimizing peak load conditions, the system reduced reliance on fossil fuels and significantly decreased greenhouse gas emissions while improving overall energy efficiency [262]. Similarly, the Heidelberg Cement Group implemented an IoT-based emission monitoring system in Germany, using sensors to track parameters like particulate matter and nitrogen oxides. This real-time monitoring enabled immediate corrective actions and optimized maintenance schedules, resulting in enhanced compliance with environmental regulations, reduced emissions, and lower operational costs [265].

In urban settings, Beijing's air quality management initiative leveraged IoT sensors and advanced analytics to monitor pollution levels and predict air quality trends. This project identified major emission sources, enabling targeted measures such as traffic restrictions and industrial controls, which contributed to significant improvements in air quality and informed urban planning efforts [266]. Likewise, in the oil and gas sector, predictive maintenance systems equipped with IoT sensors have optimized equipment performance and reduced downtime. By anticipating failures and optimizing maintenance schedules, companies like BP have minimized unplanned shutdowns, improved operational efficiency, and lowered emissions [187]. These examples demonstrate

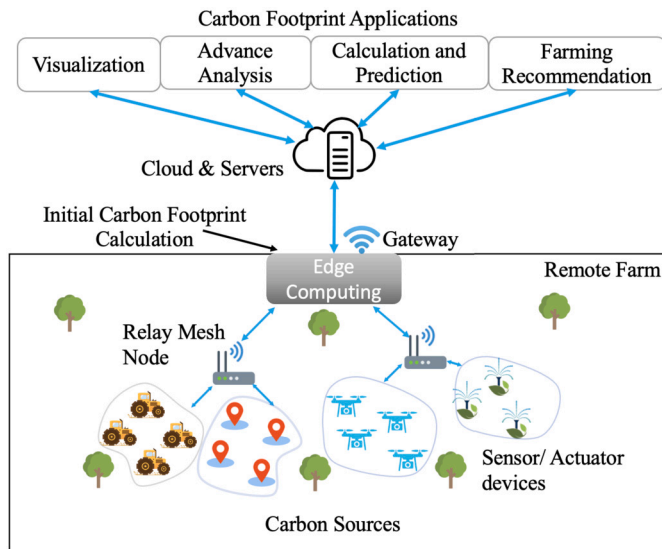


Fig. 5. An end-to-end network architecture for carbon footprint monitoring.

how IoT and Big Data technologies can address critical environmental challenges and advance sustainability across diverse industries.

Summary on Challenges and Future Research Scopes. A primary challenge in leveraging Big Data for agricultural environmental and carbon footprint monitoring is the integration of diverse data sources. Achieving real-time accuracy and managing the vast volume of data requires advanced, scalable storage and processing solutions. Transforming complex datasets into actionable insights also remains a significant task, necessitating sophisticated analytical tools. Further, balancing the environmental impact of monitoring technologies themselves is crucial.

Future research should focus on enhancing data processing algorithms for greater efficiency in handling extensive environmental datasets. Advancements in AI and ML can provide more accurate predictive models for environmental changes and carbon footprint management. Research is also needed to develop sustainable sensor technologies and eco-friendly communication networks. Additionally, formulating policies that support eco-friendly agricultural practices, underpinned by Big Data insights, will be crucial for sustainable agricultural progress. Addressing these challenges and exploring these research paths will be fundamental to evolving agricultural and environmental monitoring and reducing carbon footprints.

5. Developing end-to-end dedicated system architectures

The architecture depicted in Fig. 5 outlines an end-to-end network specifically designed for carbon footprint monitoring in agriculture. This system architecture is tailored to accurately and efficiently measure, analyze, and respond to carbon emissions associated with farming activities. Central to this architecture are various IoT “things” – sensors, actuators, and tracking devices – that play a pivotal role in sensing environmental parameters, executing automated tasks, and monitoring agricultural processes. The IoT “networks” are crucial for facilitating the seamless transfer of data to the Internet and vice versa. These networks are designed to handle the specific demands of agricultural environments, ensuring reliable and continuous data flow. As part of the application processing layer, the system employs a combination of edge computing and cloud computing methodologies alongside Big Data analytics. This multi-tiered approach allows for both real-time, on-site data processing and more extensive, cloud-based data analysis, catering to the diverse needs of carbon footprint monitoring.

Traditional carbon footprint monitoring solutions often focus on maximizing application demands, sometimes at the expense of neglecting dedicated and targeted approaches to carbon footprint monitoring.

The unique aspect of this proposed system is its “End-to-End” and “Dedicated” focus, specifically geared towards carbon footprint monitoring and sustainable farming practices. Each component of the system – from sensing devices to communication networks and computational resources – is specifically chosen and optimized for monitoring carbon emissions in agricultural settings. For instance, dedicated devices for sensing are selected based on their precision and reliability in capturing relevant environmental data, including soil moisture, temperature, GHG concentrations such as CO₂ and CH₄, solar radiation, and weather conditions. Similarly, the network infrastructure is designed to support the unique data transmission requirements of these devices, and computational resources are allocated to process and analyze the gathered data efficiently.

We propose a comprehensive system that not only addresses the technical aspects of carbon footprint monitoring but also integrates these elements into a cohesive and dedicated architecture. Such an approach ensures that every layer of the system – from data collection to processing and analysis – is fine-tuned for the specific purpose of monitoring and reducing carbon emissions in agriculture, thereby contributing to more sustainable and environmentally responsible farming practices.

5.1. System architectures leveraging IoT, Big Data, and edge computing

The development of end-to-end system architectures in agriculture, particularly for carbon footprint monitoring, involves a multi-layered approach integrating various technologies. The system architecture typically consists of the following components:

5.1.1. IoT things for data collection

Establishing an efficient IoT network for agricultural applications requires a diverse array of field equipment, encompassing both mobile and stationary devices. These devices, integral components of the IoT ecosystem, are equipped with various sensors that perform crucial functions, such as monitoring electricity usage and emissions levels and providing precise location tracking. This setup allows for real-time data collection on a wide range of environmental parameters and machinery operations.

Table 4 categorizes a diverse range of IoT-compatible sensors essential for monitoring carbon and greenhouse gas emissions in agriculture. These sensors play a pivotal role in optimizing irrigation, analyzing climate patterns, assessing crop health, and monitoring environmental conditions, collectively facilitating a comprehensive approach to sustainable farming practices. Exhaust gas analyzers [151,136] are essential for monitoring machinery emissions, while power meters track the energy consumption of agricultural equipment [140]. Similarly, GPS trackers integrated into mobile agricultural machinery provide accurate location data, enhancing operational planning and efficiency (e.g., [170] use GPS in IoT-based connected vehicles). Additionally, sensors like soil moisture sensors and NDVI sensors work together to optimize irrigation and evaluate crop health, minimizing resource wastage and improving sustainability. Other critical sensors include weather stations, air quality monitors, and infrared gas analyzers, which offer insights into meteorological data, greenhouse gas levels, and environmental conditions. The variety of sensors listed in Table 4 underscores the robust data collection capabilities of IoT networks, enabling accurate monitoring and informed agricultural decision-making for effective carbon footprint management.

5.1.2. IoT networks for data collection

The IoT Network is pivotal in precision agriculture, forming the backbone of data collection. It relies on a variety of robust communication and networking standards and protocols to ensure seamless and efficient data flow. A key consideration in these networks is supporting dedicated bandwidth or channels for specific communication protocols.

Table 5 categorizes various communication network technologies essential for IoT-driven carbon footprint monitoring in agriculture. These

Table 4

Categorized Sensors for Monitoring Carbon and Greenhouse Gases in Agriculture.

Sensor Type	Purpose	References
Emission Monitoring Sensors		
Exhaust Gas Analyzers	Measure emissions from combustion engines used in agricultural machinery, including CO ₂ , NO _x , and other pollutants, to optimize fuel efficiency and emission control.	[151,136,89,136]
Infrared Gas Analyzers	Detect and quantify greenhouse gases such as CO ₂ , CH ₄ , and N ₂ O in agricultural settings, supporting precision management of emissions.	[173,150,77]
Air Quality Monitors	Measure ambient concentrations of key pollutants and GHGs like CO ₂ and methane, helping monitor air quality and assess the impact of agricultural activities.	[253,168,58,219]
Environmental and Meteorological Monitoring Sensors		
Weather Stations	Record and analyze weather parameters such as temperature, humidity, rainfall, wind speed, and solar radiation to assess climate conditions affecting crop growth.	[7,72,240]
GPS Trackers	Provide precise geolocation for mobile agricultural equipment, enabling field mapping, logistics management, and accurate tracking of emission sources.	[31,166,170]
Soil and Crop Monitoring Sensors		
Soil Moisture Sensors	Continuously monitor soil moisture levels to optimize irrigation schedules, improve water use efficiency, and minimize water-related emissions.	[164,85,68]
NDVI Sensors	Measure vegetation health and density by detecting changes in plant reflectance, aiding in the early detection of stress and optimized resource allocation.	[144,40,120]
Energy and Power Consumption Monitoring Sensors		
Power Meters	Monitor electricity consumption of agricultural equipment such as irrigation pumps and grain dryers, enabling energy-efficient operation and cost savings.	[51,140]

Table 5

Network Technologies in IoT for Carbon Footprint Monitoring and Sustainable Agriculture.

Network Type	Description	References
Unlicensed Spectrum Networks		
Wi-Fi Networks	High-bandwidth wireless networks suitable for short-distance communication, typically used in stationary or limited-range mobile IoT devices.	[152,94,180]
Wi-Fi HaLow	Extended-range Wi-Fi technology with dynamic bandwidth allocation, enabling connectivity for a large number of IoT devices in remote areas.	[10,159,81]
Zigbee/6LoWPAN Networks	Low-power, short-range communication protocol ideal for connecting low-data-rate sensors within local farm networks	[238,191,129]
LoRaWAN	Long-range, low-power wireless network technology operating in unlicensed spectrum bands, ideal for monitoring remote fields and sensors.	[30,141,22]
Licensed Spectrum Networks		
Cellular Networks (4G/5G)	Long-range networks with extensive coverage, useful for remote monitoring and real-time data transfer from agricultural machinery and mobile devices.	[21,156,199]
NB-IoT/SigFox	Narrowband IoT technology optimized for low power consumption and long-range communication in agricultural applications such as soil moisture and livestock monitoring.	[185,35,167]
Hybrid and Customizable Network Solutions		
Hybrid Approaches	Combination of technologies like Wi-Fi, 6LoWPAN, LoRa and Cellular to maximize coverage, redundancy, and efficiency in large-scale smart farming.	[9,7,172]

networks, operating across licensed and unlicensed spectrums, facilitate seamless data transmission from sensors and IoT devices, ensuring real-time monitoring and adaptive decision-making for sustainable agricultural practices. It emphasizes the need to balance range, data rate, and power efficiency in network architecture. Current network architectures in precision agriculture primarily rely on wireless sensor networks, utilizing a range of communication technologies such as ZigBee [108], Bluetooth [12], cellular networks including Fourth Generation (4G) and emerging Fifth Generation (5G) technologies [239], Wi-Fi [9], and LoRa [117]. However, managing these networks, particularly in modern and future farm cyber-physical systems, presents significant challenges.

The table outlines the types of networks used, their specific applications, and references to studies that have implemented or explored these technologies. It underscores the diverse needs of IoT networks in agriculture, from high-bandwidth Wi-Fi for stationary devices to long-range cellular and specialized IoT protocols for extensive agricultural fields. Hybrid approaches are often employed to maximize efficiency and coverage, ensuring continuous data transmission. The emerging potential of

Wi-Fi HaLow [159], in particular, shows promise for its dynamic bandwidth and long-range support, making it suitable for connecting a large number of IoT devices in agricultural settings.

Advanced network architectures play a pivotal role in precision agriculture, particularly in the realm of carbon footprint monitoring. These architectures incorporate cutting-edge techniques such as Network Slicing [256], Network Functions Virtualization (NFV) and Software-Defined Networking (SDN) [202], etc. These technologies are fundamental in allocating dedicated resources tailored to carbon monitoring applications [59,228]. Network slicing, for instance, allows for the creation of separate, virtualized network partitions, each optimized for a particular service or application. This is especially beneficial in ensuring that carbon monitoring applications have the bandwidth and latency characteristics they need without interference from other network traffic.

NFV and SDN bring additional layers of flexibility and efficiency to network management. NFV focuses on abstracting network functions from hardware, allowing for more agile and cost-effective network ser-

Table 6
Categorized Edge, Fog, and Dew Computing Solutions for Carbon Footprint Monitoring and Sustainability in Agriculture.

Work	Description and Example Use Cases	References
Edge Computing Solutions		
Edge Computing in Agriculture	Real-time data processing at the field level to support efficient decision-making in resource management, pest detection, and emissions reduction.	[123,258,218]
Edge Analytics for Carbon Monitoring	Integration of sensors and IoT devices with localized processing units to measure and mitigate carbon emissions in agricultural and smart environment applications.	[225,179,106]
Fog Computing Solutions		
Fog Computing for Emissions Tracking	Intermediate data aggregation and processing between edge devices and the cloud, enabling efficient real-time tracking and analysis of emissions in agricultural operations.	[15,130,192]
Collaborative Fog Networks	Multi-tier fog networks used to coordinate farming activities and optimize emissions reduction strategies across farms and regions through real-time data exchange and analysis.	[9,197,106]
Dew Computing Solutions		
Dew Computing in Environmental Monitoring	Decentralized data processing performed directly at the data source (e.g., sensors and IoT devices) for efficient environmental monitoring without the need for continuous cloud connectivity.	[128,41]

vices. SDN, on the other hand, centralizes network control, making it easier to dynamically allocate resources and optimize network performance for specific tasks like carbon data transmission. This level of control is crucial for maintaining the quality of service required by carbon monitoring applications, ensuring that data is transmitted reliably and efficiently. The role of Radio Access Network (RAN) in this setup is also significant, particularly in terms of handling the wireless communication aspects [194]. RAN’s ability to manage radio resources effectively ensures that IoT devices deployed in the field for carbon monitoring can communicate their data back to central servers without loss or delay. Together, these advanced network architectures provide a robust foundation for dedicated carbon monitoring applications in agriculture. They ensure that the necessary computational and data transmission resources are available, reliable, and optimized for the specific requirements of carbon footprint assessment, thereby enhancing the precision and effectiveness of sustainable farming practices.

5.1.3. Edge-based carbon footprint calculation

Implementing carbon footprint calculations at the edge of the network is pivotal for achieving low-latency decision-making and efficient data transmission in agriculture settings. This approach utilizes GWP factors and real-time data processing to accurately quantify emissions from agricultural activities. By categorizing these emissions into three tiers—direct emissions (Scope 1), emissions related to purchased energy (Scope 2), and all other indirect emissions (Scope 3)—a comprehensive view of the agricultural carbon footprint is obtained.

The adoption of edge, fog, and dew computing solutions in agricultural carbon footprint monitoring enables real-time analytics, reduces dependency on centralized cloud infrastructure, and enhances energy efficiency. These computing paradigms facilitate localized data processing, empowering farmers with actionable insights while minimizing network congestion and response latency (see Table 6). For example, edge computing is utilized for real-time data processing in agricultural settings, significantly reducing latency and enhancing the responsiveness of emission tracking systems [123]. Fog computing, as demonstrated in the table, plays a critical role in tracking and analyzing emissions data, offering a midway computation solution between edge devices and cloud servers [216]. Dew computing, an emerging field, is applied for decentralized data processing in environmental monitoring, highlighting its utility in local data processing without relying on centralized cloud services [128]. Edge analytics are increasingly being employed to quantify carbon emissions directly in the agricultural fields, integrating sophisticated data analysis techniques in the real-time monitoring process [225].

This localized processing capability ensures that carbon footprint calculations are executed with heightened accuracy and timeliness. Such precision is crucial in significantly reducing the environmental impact

of agricultural activities, however the implementation of these technologies in agriculture is challenging. Future research needs to focus on aspects such as dedicated service requirements for the proposed systems, efficient allocation of resources, and collaborative computing approaches. For instance, Karl et al. [221] have explored the potential of managing GHG emissions using edge computing through a collaborative approach. This demonstrates how joint efforts in data processing can enhance the effectiveness of environmental monitoring. Additionally, the concept of reconfigurable edge computing, as suggested in recent studies [13], opens new avenues for dynamic task allocation and resource management in edge computing architectures. This flexibility is particularly beneficial in agricultural settings where varying data loads and processing needs require adaptable and scalable solutions. By addressing these areas, future advancements in edge computing can significantly bolster the precision and efficacy of carbon footprint monitoring in agriculture, further aligning farming practices with environmental sustainability goals.

5.1.4. AI-enabled cloud-based models

Developing cloud-based models powered by AI is crucial for comprehensively analyzing the vast amounts of data collected from agriculture settings. These models utilize advanced Machine Learning (ML) and deep learning techniques to predict and optimize carbon footprints [103,223,131]. They analyze a wide range of parameters, including mobility patterns, engine sizes, and specific farm activities, to provide actionable insights for sustainable agricultural practices.

The integration of AI-powered cloud computing, collaborative learning, and advanced analytical models in agriculture enables precise carbon footprint monitoring and enhances sustainability. By leveraging scalable computing platforms, federated AI, and reinforcement learning, these approaches provide data-driven insights for optimizing resource use, mitigating emissions, and ensuring resilient farming practices (see Table 7). For example, federated learning offers a decentralized ML approach, enhancing data privacy in sustainable agriculture [38]. Graph-based models are used to analyze complex relationships within agricultural data, leveraging graph theory for deeper insights [119]. Reinforcement learning, known for its adaptive decision-making capabilities, is employed to optimize farming practices [90]. The integration of these advanced AI models into cloud computing frameworks represents a significant leap in climate-smart agriculture, enabling more precise and predictive insights into environmental impacts and facilitating smarter, data-driven decisions for sustainable farming practices. They offer scalable, dynamic, and sophisticated solutions for processing complex agricultural data. This facilitates real-time decision-making and significantly contributes to sustainable farming practices, aligning agricultural operations with broader environmental and sustainability goals.

Table 7
Categorized AI-Enabled Cloud-Based Models for Carbon Footprint Monitoring and Sustainability.

Technique	Description and Example Use Cases	References
Data Processing and Analysis Techniques		
Scalable Cloud Computing	Cloud-based platforms such as AWS are used for large-scale agricultural data processing, including plant recognition through image data and real-time decision-making for optimized resource use.	[91,237,66,67]
Edge ML	Enables real-time, on-field pest detection and localized, energy-efficient neural network partitioning for carbon-aware agricultural monitoring.	[258,79,132]
Collaborative and Privacy-Preserving AI		
Collaborative AI	Supports cooperative and collaborative AI models for sustainable agriculture by facilitating real-time data sharing and coordination across farms and regions.	[97,70,213,251]
Federated Learning	Uses decentralized ML for sustainable agriculture while ensuring data privacy and security through distributed analytics.	[38,251,143]
Advanced Models for Complex Agricultural Relationships		
Graph-Based Models	Analyzes complex agricultural systems using graph theory and graph neural networks to model GHG emissions and predict crop yields for better decision support.	[119,96,71]
Reinforcement Learning	Adaptive decision-making models optimize resource management, such as irrigation schedules and fertilizer application, to reduce GHG emissions and enhance sustainability.	[49,90,41]

5.1.5. User interface and automation for sustainable farming recommendations

Developing a user-friendly and intuitive graphical user interface (GUI) is pivotal for providing farmers and administrators with actionable insights and recommendations for sustainable farming practices [101,204]. This GUI serves as a crucial tool in translating complex data analyses into easily understandable and actionable information. It enables users to make informed decisions to reduce carbon emissions and foster sustainable agricultural methods. The GUI is not just a static display of information but an interactive platform that facilitates real-time decision-making. It should be capable of automatically adjusting and reconfiguring based on incoming data, thereby providing the most relevant and up-to-date information to the users. For instance, the interface could display real-time carbon emission levels, suggest optimal irrigation schedules, or alert about potential overuse of fertilizers based on the data processed by AI-enabled models.

Further, the interface should be integrated with systems that can relay information directly to remote field operations. This integration ensures that the recommendations are not just theoretical but are translated into practical actions. For example, based on the GUI's insights, automated systems could adjust irrigation systems, modify the application of fertilizers, or reconfigure machinery settings to optimize for lower carbon emissions [205,50]. These adjustments could occur both in physical operations (like machinery control) and in computational aspects (such as data processing parameters) and communication (adjusting IoT device reporting frequency). The GUI should also support remote reconfiguration capabilities, allowing farm administrators to make changes to IoT device settings or farming strategies from a distance. This feature is particularly useful in large-scale farming operations where real-time adjustments across vast areas can lead to significant improvements in sustainability.

6. Identifying challenges and research opportunities

Integrating IoT, Big Data, Edge Computing, and ML into a dedicated, end-to-end system presents unique challenges. Yet, it also opens up numerous opportunities for enhancing the sustainability and efficiency of agricultural practices. This section delves into these aspects, providing insights into the hurdles faced in real-world applications and the potential avenues for future advancements.

One of the primary challenges is the complex system integration required to harmonize diverse technologies like IoT, Big Data, and Edge Computing into a cohesive framework [250]. Ensuring seamless interoperability, efficient data flow, and real-time processing across various system layers poses a significant technical challenge. Further, managing the vast volume and variety of data generated by IoT devices for carbon

footprint monitoring is a daunting task. This challenge calls for advanced Big Data analytics and ML algorithms that can efficiently store, process, and extract meaningful insights from this data. Another major challenge lies in the domain of energy efficiency and sustainability. While IoT and Edge Computing devices are essential for real-time monitoring and data collection [123], they increase the energy footprint of agricultural practices. Developing energy-efficient devices and sustainable system architectures is crucial to ensure that these technologies contribute to an overall reduction in the carbon footprint.

The accuracy and reliability of predictive models powered by ML also pose a challenge [267], especially considering changing environmental conditions and varied agricultural practices. Ensuring that these models can reliably predict carbon emissions and other environmental impacts under diverse conditions is essential for their effectiveness. Scalability and adaptability are also critical factors. The system must be scalable to accommodate different sizes and types of agricultural operations and be adaptable to a range of environmental conditions and farming practices. This requirement makes its deployment challenging in diverse agricultural settings. On the flip side, these challenges open up numerous opportunities. Advancements in AI and ML offer the potential for more accurate and efficient analysis of environmental data, including the development of adaptive models that can predict carbon emissions under varying conditions. The field of sensor technology also presents opportunities for innovation, with the development of more energy-efficient sensors capable of capturing a wider range of environmental data, significantly enhancing carbon footprint monitoring.

The synergy between cloud and edge computing offers another avenue for improvement [134]. This synergy can balance computational loads, improve system efficiency, and enable real-time decision-making through local data processing at the edge, complemented by more complex, large-scale data analyses in the cloud. User-centric design in interfaces and decision-support systems can make these technologies more accessible and practical for farmers. Intuitive GUIs and automated systems directly translating data insights into sustainable practices can significantly aid in their adoption.

Another promising avenue is the integration of blockchain technology into carbon credit trading systems. Blockchain provides a transparent, tamper-proof ledger that ensures the authenticity and traceability of carbon credits generated through sustainable agricultural practices [32]. By recording data such as emission reductions and carbon sequestration metrics from IoT-enabled systems directly onto a blockchain, farmers can demonstrate compliance with environmental standards and generate verifiable credits [63]. These credits can be traded in global carbon markets, providing farmers with an additional income stream while incentivizing the adoption of sustainable practices. Moreover, blockchain can facilitate decentralized carbon markets, reducing the

role of intermediaries and ensuring fairer compensation for farmers. This approach not only boosts economic opportunities but also addresses challenges related to data integrity and market trust in carbon offset mechanisms, thus advancing the scalability and credibility of carbon credit systems in agriculture.

Collaboration with policymakers to develop supportive frameworks and incentives for the adoption of advanced carbon footprint monitoring systems is also a crucial opportunity. Such collaboration can drive innovation and foster the wider adoption of sustainable agricultural practices. Moreover, there is an opportunity to develop solutions that address not just technological challenges but also consider environmental, economic, and social sustainability in a holistic manner. This approach can lead to more robust and widely accepted solutions in the agricultural sector. Addressing these challenges and harnessing these opportunities will be key to advancing sustainable agriculture in the coming years.

7. Conclusion

In conclusion, this review demonstrates that the synergy of IoT, Big Data, and AI shows strong potential for reducing agriculture's carbon footprint and facilitating sustainable resource use. By adopting a Scope-based methodology in line with the GHG Protocol, we have highlighted how direct (Scope 1), indirect (Scope 2), and other indirect (Scope 3) emissions can each be monitored, analyzed, and addressed through well-coordinated digital systems. Recent case studies and scholarly work underscore real-world scenarios where sensors, connectivity, and ML applications have been successfully deployed to optimize fertilizer usage, enhance irrigation scheduling, and improve livestock management—each contributing to lowered emissions and better resource efficiency.

Despite these advances, challenges remain. Infrastructure limitations, high implementation costs, and data privacy concerns must be resolved to promote equitable access and widespread adoption. Standardizing protocols for interoperability and quality control is also critical to ensure accurate measurement and reporting of emissions. Future research should further refine ML models to handle local variability in climate, soils, and farm practices while exploring collaborative frameworks (e.g., federated learning) that protect data privacy. Moreover, integrating policy incentives, capacity-building programs, and stakeholder collaborations can accelerate the use of IoT and Big Data tools in real-world agricultural settings. By addressing these issues, the agricultural sector can leverage digital innovations to reduce carbon footprints, foster resilience to climate change, and advance global sustainability goals.

CRedit authorship contribution statement

Nurzaman Ahmed: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Nadia Shakoor:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

N.S. is a co-founder of Agrela Ecosystems, a company specializing in IoT-based sensing technologies and data-driven solutions for precision agriculture and environmental monitoring. The other authors have no conflict of interest to declare.

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Data availability

No data was used for the research described in the article.

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